

ABSTRACT

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This work presents a comprehensive study on Next-Generation Wireless Electric Vehicle (EV) Charging systems, which employ electromagnetic energy transfer techniques to enable cable-free power transmission between charging infrastructure and electric vehicles. The proposed system primarily relies on Inductive Coupling and Resonant Magnetic Coupling principles, wherein an alternating magnetic field generated by a ground-based transmitter coil induces an electromotive force (EMF) in a receiver coil mounted beneath the vehicle, thereby facilitating efficient energy transfer. Conventional plug-in charging systems suffer from critical limitations such as mechanical wear of connectors, user inconvenience, exposure to high-voltage cables, limited charging infrastructure, and increased maintenance costs. To mitigate these challenges, wireless charging technology integrates embedded charging pads within parking spaces and roadways, enabling static and dynamic charging capabilities. The developed framework supports dynamic wireless power transfer (DWPT), allowing EVs to charge while in motion through smart road infrastructure embedded with transmitter coils. This approach significantly reduces range anxiety, optimizes battery capacity requirements, and enhances overall energy efficiency in urban mobility systems. From an application perspective, wireless charging technology demonstrates substantial potential in commercial transportation, including delivery fleets, public buses, and autonomous electric vehicles, by minimizing downtime and enabling continuous energy replenishment. Furthermore, the system improves safety by eliminating exposed conductive components and reducing risks associated with electrical hazards. Despite its advantages, several technical and regulatory challenges remain, including power transfer efficiency, electromagnetic interference (EMI), standardization of charging protocols, interoperability between manufacturers, and compliance with global safety regulations. Ongoing pilot implementations in regions such as London, Sweden, Germany, and the United States validate the feasibility and scalability of this technology. In conclusion, wireless EV charging represents a transformative advancement in sustainable transportation, paving the way for a fully automated, energy-efficient, and wire-free electric mobility ecosystem.